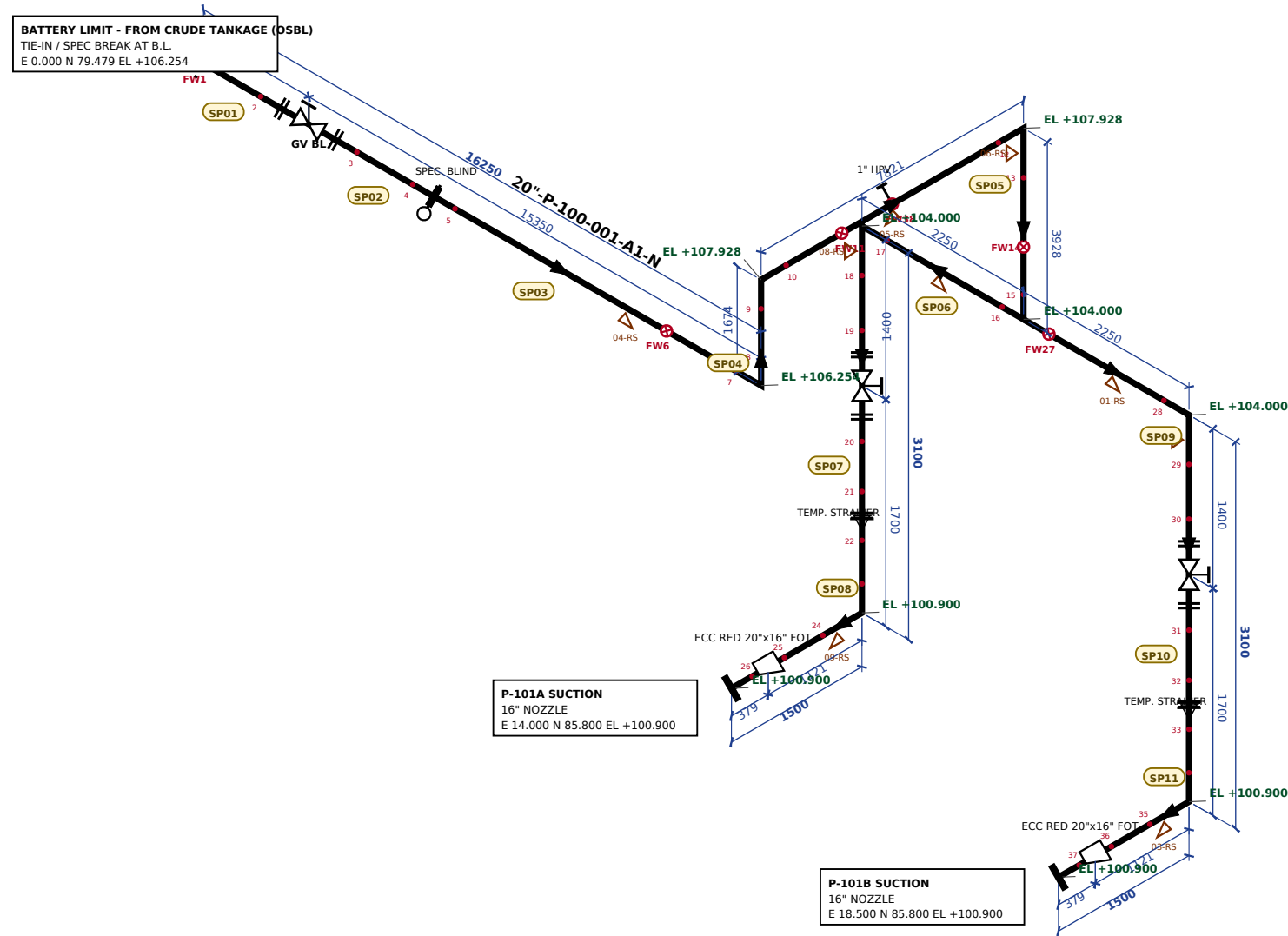
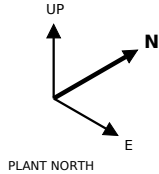


ISOMETRIC VIEW - NOT TO SCALE - DIMENSIONS IN mm, ELEVATIONS / COORDINATES IN m (PLANT GRID)



BILL OF MATERIALS (THIS SHEET)

PT	DESCRIPTION	NPS	QTY
1	PIPE STD, BE, ASTM A106 Gr.B SMLS	20"	27.2 m
2	FLANGE WN RF CL150 B16.5, bore STD, ASTM A105	20"	12
3	ELBOW 90 LR, BW, STD, ASTM A234 WPB	20"	7
4	TEE EQUAL, BW, STD, ASTM A234 WPB	20"	1
5	REDUCER ECC, BW, STD, ASTM A234 WPB	20" x 16"	2
6	FLANGE WN RF CL150 B16.5, bore STD, ASTM A105	16"	2
7	SOCKOLET 3000# (branch), ASTM A105	1"	1
8	ERECTION: GATE VALVE OS&Y CL150 RF, A216 WCB, trim 8	20"	3
9	SPECTACLE BLIND CL150, ASTM A105	20"	1
10	TEMPORARY CONE STRAINER (commissioning)	20"	2
11	GASKET SPW 304/graphite, outer ring, CL150 B16.20	20"	12
12	STUD BOLT SET (nuts) A193 B7 / A194 2H, CL150	20"	12
13	PIPE SUPPORTS (std. dwgs): rest x9	-	9

CUT LENGTHS (mm, ex. weld gap) / SPOOLS

PIECE	NPS	L	PIECE	NPS	L	PIECE	NPS	L
SP01-1	20"	528	SP02-1	20"	485	SP03-1	20"	11250
SP04-1	20"	2195	SP04-2	20"	150	SP04-3	20"	1650
SP05-1	20"	4647	SP05-2	20"	1650	SP06-1	20"	1135
SP06-2	20"	1107	SP06-3	20"	266	SP07-1	20"	186
SP08-1	20"	95	SP08-2	20"	105	SP09-1	20"	1107
SP09-2	20"	266	SP10-1	20"	186	SP11-1	20"	95
SP11-2	20"	105						

SPOOL DEVELOPED LENGTHS (transport limit 12.0 x 3.0 x 3.0 m):

SP01=0.7m SP02=0.8m SP03=11.4m SP04=7.0m SP05=7.8m SP06=4.9m SP07=0.5m SP08=2.5m SP09=3.0m SP10=0.5m SP11=2.5m

NOTES

- Pipe/fittings/flanges per class A1 (CFU-000-PI-SPC-001); BW fittings ASME B16.9, flanges B16.5/B16.47A RF.
- Hydrotest 5.6 barg per ASME B31.3 345.4 (vents at high points, drains at low points).
- PWHT: NO. NDE: 10 % RT of BW.
- Supports: RS rest/shoe, GD guide, ANC anchor, SPR variable spring - tags PS-<area><seq>-nn.
- Welds: dot = shop weld, crossed circle FW = field weld; spools <= 12 m for transport.
- Pump suction: no pockets, header falls to pumps; eccentric reducers 20"x16" FLAT ON TOP; temporary strain
- Pump nozzle 16" assumed (API 610 vendor data pending).
- Rack high point vented; tank static head gives NPSH margin (check at detailed design).
- Header at y = pump nozzle + 1.5 m limited by E-106A/B (layout) - see RPT-001 issues.

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT		
PIPING ISOMETRIC 20"-P-100-001-A1-N CRUDE SUCTION TO P-101A/B		
DISCIPLINE PIPING	STAGE FEED	SCALE NTS
SHEET 1 OF 1		
DRAWN CLAUDE CODE	CHECKED -	APPROVED -
DATE 2026-10-02		
DRAWING NUMBER CFU-100-PI-ISO-005		REV A

LINE NUMBER 20"-P-100-001-A1-N	FROM TK (OSBL)	TO P-101A/B	PIPING CLASS A1 - 150# CS ASTM A106 Gr.B
DESIGN P / T 3.7 barg / 60.0 C	OPER. P / T 1.99 barg / 30 C	TEST P (HYDRO) 5.6 barg (1.5xPxST/S=1.0)	PIPE WALL STD (9.53 mm)
INSULATION N 0 mm	PWHT NO	NDE 10 % RT of BW	STRESS CAT 1 - rotating equipment (API 610 nozzle loads)
P&ID CFU-100-PR-PID-001	FLUID / PHASE P / L Crude from tankage to ch	PAINT Epoxy/PU system	SPOOLS / WELDS 11 spools; 32 shop / 6 field